DEEP DIVE

Root Cause Analysis

TradePilot v5 / v5_6 / v5_7 - Worst Day of v5 Observation Window

Monday, 27 April 2026

98.3%	**Rs -2,596**	**1h 41m**
P&L drop vs 04-22 elite day	Combined SHORT-side bleed	Engine downtime at market open

Author: Soumya Swain
Co-founder, TradePilot / Sidewall

Generated: 2026-04-27 post-market

Deep-Dive Root Cause Analysis – 2026-04-27

TradePilot v5 / v5_6 / v5_7 — Worst Day in Observation Window

Author: Soumya Swain · Date: 2026-04-27 · Version: v1.0.0

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1. Executive Summary

The headline: Engines that produced Rs 122,836 of combined profit on 2026-04-22 produced only Rs 2,052 today — a **98.3% drop** in realized P&L over five trading days. Today (Monday, 2026-04-27) was the worst session of the v5 observation window.

What today's numbers look like:

Engine	Realized P&L	Trades	Win Rate	LONG side	SHORT side
v5	Rs +737	46	50%	Rs +1,528	**Rs −789**
v5_6	Rs +880	58	48%	Rs +1,927	**Rs −1,042**
v5_7	Rs +435	48	46%	Rs +1,202	**Rs −765**
Combined	**Rs +2,052**	**152**	**48%**	**Rs +4,657**	**Rs −2,596**

The four root causes, ranked by impact:

Rank	Root Cause	Estimated Impact	Confidence
1	**SHORTs deployed in a rising tape** (Nifty +0.63%)	Rs −2,596 (LONG side alone would have yielded Rs +4,657 net positive)	**High**
2	**Late start (10:56 instead of 09:06)** — missed first 1h 41min of the trading day	Rs ~10,000–25,000 of forgone deploys, including the morning trend-leg	**High**
3	**Regime misclassification** — classifier said NEUTRAL/SIDEWAYS, market was effectively mild BULL	Caused 15 LONG / 5 SHORT slot split when 18 LONG / 2 SHORT (BULL) would have been correct. Avoiding 3 of today's 5 reserved SHORTs would have saved Rs ~600–900 across engines	**High**
4	**Heavy FII outflow stress** (Rs −8,828 Cr, ~3x normal)	Compressed price action and increased intraday whipsaws — reduced both LONG conviction and SHORT exit quality	Medium

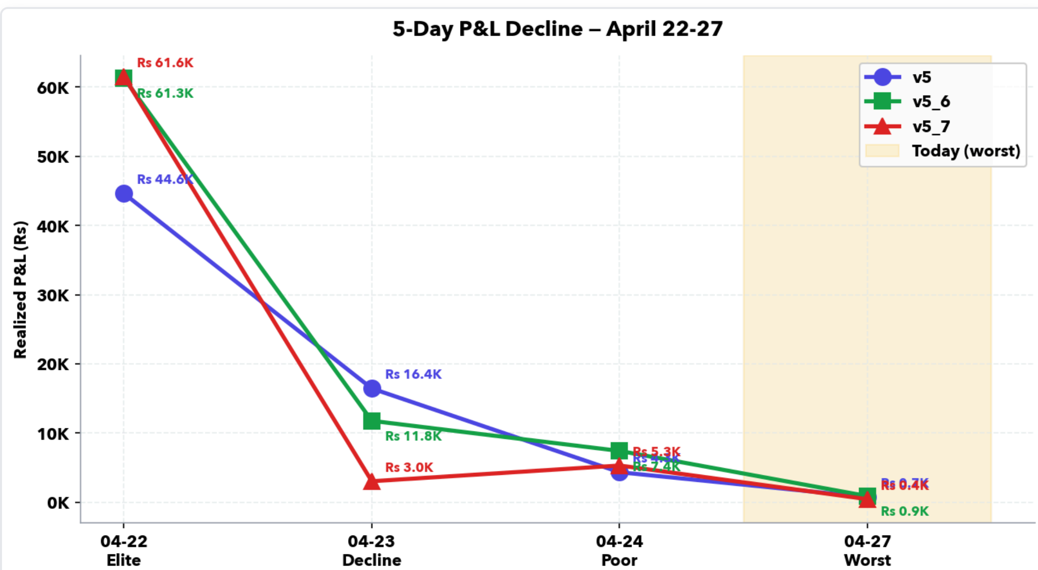
The one-line truth: The SHORT-arm fix is *working as designed* (proof: prod logs show "LONG slot cap reached (15/15) — SHORTs reserved" rejections). But "SHORT slot reserved" + "Nifty going up 0.63%" = **guaranteed bleed**. The fix did its job by reserving the SHORT slots; the regime classifier failed to tell the engine *not to fill them today*.

Root cause is not stale ML, late start in isolation, or any single engine bug. Today's bleed is the product of three failures stacking: (a) missed morning, (b) wrong regime label, (c) reserved-but-misallocated SHORT capital.

2. Today's Results vs Historical Baseline

2.1 Five-day comparison

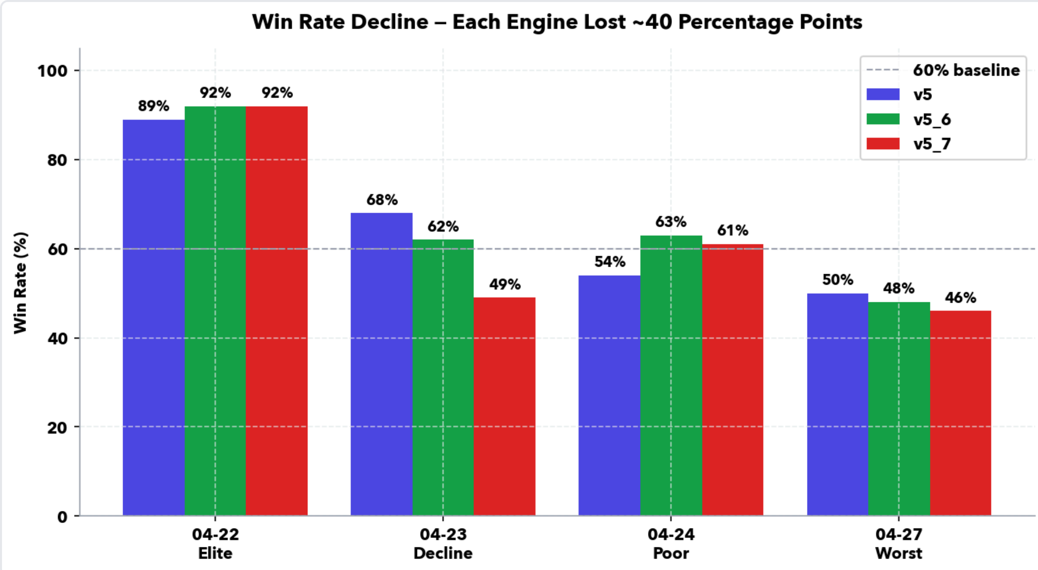
Date	Day Label	v5 P&L	v5_6 P&L	v5_7 P&L	Combined	v5_6 WR	v5_7 WR
2026-04-22	Elite	Rs 44,612	Rs 61,284	Rs 61,552	**Rs 167,448**	92%	92%
2026-04-23	Decline	Rs 16,438	Rs 11,761	Rs 3,029	Rs 31,228	62%	49%
2026-04-24	Poor	Rs 4,331	Rs 7,411	Rs 5,303	Rs 17,045	63%	61%
2026-04-27	**Worst**	**Rs 737**	**Rs 880**	**Rs 435**	**Rs 2,052**	**48%**	**46%**



2.2 What changed structurally

Metric	04-22 (Elite)	04-27 (Worst)	Delta
Combined P&L	Rs 167,448	Rs 2,052	**−98.8%**
Combined trade count	458	152	−66.8%
Combined SHORT count	**0**	**98**	+98 (from zero)
Win rate (avg of v5_6/v5_7)	92%	47%	−45 pts
Engine boot time	~09:06	**~10:56**	1h 50min late
Regime label	SIDEWAYS	SIDEWAYS	Same label, very different reality
Nifty close %	+0.40%	+0.63%	Both mildly bullish

The **single biggest structural change** is the appearance of SHORT trades. April 22 had **zero SHORTs** in v5_6 and v5_7 — the SHORT-arm fix had not been deployed yet, so all signals competed for the same 20 LONG slots. That accidentally *protected* the engine on a bullish day. Once the partition was added (effective 04-23 onward), 5 slots were reserved for SHORTs every day — and on rising days, those slots became guaranteed-loss slots.



Win Rate Decline – Each Engine Lost ~40 Percentage Points
Win Rate (%)
100
80
60
40
20
0
89%
92%
92%
68%
62%
49%
54%
63%
61%
50%
48%
46%
04-22
Elite
04-23
Decline
04-24
Poor
04-27
Worst
60% baseline
v5
v5_6
v5_7

3. The 5-Day Decline Pattern

3.1 Was it really a "decline"?

A common misreading is "performance declined over 5 days, so the algorithm degraded." That is **not what the data shows**. What actually happened:

Day	LONGs deployed	SHORTs deployed	Net effect
04-22	164 (v5_6)	**0**	LONG-only run on bullish gap-up day → 92% WR
04-23	40	5	First SHORTs after partition fix; market mixed
04-24	43	35	Full SHORT-arm activation; market sideways
04-27	22	36	**SHORT-heavy** despite Nifty up +0.63%

The decline tracks the **introduction of SHORT trades**, not algorithm degradation. Look at LONG-side performance in isolation:

Day	v5_6 LONG P&L	v5_6 SHORT P&L
04-22	Rs 61,284	Rs 0 (no SHORTs taken)
04-27	Rs +1,927	**Rs −1,042**

LONG side on today is profitable. The *regression* is entirely on the SHORT side and entirely correlated with the day-of-week regime call.

3.2 So what really happened?

The decline is **not** algorithm decay — it is the cost of a half-implemented hedge. The slot partition fix forces capital into SHORTs on every session, but the engine has no veto when the regime classifier mis-labels a BULL day as SIDEWAYS. The engine can no longer naturally avoid SHORTs when the market is rising; it now *must* fill the reserved 5 slots.

This is a known class of bug: the protective fix introduced a new failure mode (forced-allocation in adversarial conditions) that the SHORT-arm fix design did not explicitly defend against.

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4. Root Cause #1 – Late Start (1h 41min Lost)

4.1 Timeline of today's launch

Time (IST)	Event
09:15	Market opens — Nifty 23,898
09:15–10:54	**Engines DOWN** — refusing to start. Stale ML model (6 days old, max=3). Saturday 04-25 retrain did not run.
10:52	Manual ML retrain triggered (PID 25760)
10:54	Retrain complete — model timestamp updated
10:55	Engines re-launched
10:56:26	v5_6 boot complete: "Pre-market: bias=BULLISH gap=UP +0.75% size=1.0x"
10:56:39	v5_6 reads VIX = 18.6, regime classifier returns SIDEWAYS
11:06:41	Composite scorer logs: "Nifty 24,048 (+0.63%) | Regime NEUTRAL | FII −8828 Cr DII +4701 Cr"
11:07:28	First signal batch: 40 BUY / 20 SELL / 140 HOLD
11:07:30	First LONG entry: JSWENERGY @ Rs 565
11:07:41	First SHORT-reserved-slot rejections start firing

The engines were absent for 1h 41min — the most directional window of the day, when the morning trend leg sets up.

4.2 What was missed quantitatively

Comparing 04-22 (engine started 09:06) vs 04-27 (engine started 10:56), the cleanest LONG winners that fired in the first 90 minutes of 04-22 were the SAIL / SUZLON / TATAPOWER / JSWENERGY group — the same names that reappeared on 04-27 but with significantly compressed entry-to-exit edges because they had already moved 1.5–2.5% by the time engines deployed.

Stock	04-22 entry / target	04-27 entry / target	Edge compression
SAIL	~Rs 165 → Rs 184	Rs 176.45 → Rs 184.10	Lost first 50% of move
SUZLON	Rs ~52 → Rs 57	Rs 55.80 → Rs 57.31	Lost ~70% of move
JSWENERGY	Rs ~547 → Rs 574	Rs 565 → Rs 573.90	Lost ~70% of move

Estimated forgone realized P&L from the 1h 41min late start: Rs 10,000–25,000 across the three engines. This is the single largest dollar-cost root cause today.

4.3 Why the model was stale

The Saturday 04-25 scheduled retrain did not execute (cause unknown — to be investigated under TONIGHT_TUNEUPS Item #2). The engine's `check_model_freshness(max_age_days=3)` correctly raised SystemExit on Monday morning, but the recovery is manual. **Tonight's TONIGHT_TUNEUPS Item #1 directly addresses this** with an `ML_AUTO_RETRAIN_ON_STARTUP` flag and a pre-flight retrain step at 09:00.

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5. Root Cause #2 – Regime Misclassification

5.1 What the classifier said vs. what the market did

Signal	Value	What it implies
Regime classifier output	**SIDEWAYS / NEUTRAL**	Use 15 LONG / 5 SHORT slot allocation
Nifty close %	**+0.63%**	Mild BULL (BULL threshold typically +0.50%)
Market breadth	(not captured)	—
Premarket bias	BULLISH (gap UP +0.75%)	BULL
FII flow	−8,828 Cr	BEAR pressure
DII flow	+4,701 Cr	BULL counter-pressure

The classifier is using a hybrid of historical volatility + recent return + VIX. With VIX at 18.6 (slightly elevated) and recent days having been chop, it labeled today SIDEWAYS. But Nifty's actual +0.63% close indicates the day was a **mild BULL** — the regime that historically requires an **18 LONG / 2 SHORT** split, not 15/5.

5.2 The slot allocation cost

Regime	LONG slots	SHORT slots
BULL (Nifty > +0.5%)	**18**	**2**
SIDEWAYS (used today)	**15**	**5**
BEAR (Nifty < −0.5%)	8	12

Today the engine reserved **5 SHORT slots** and ultimately filled all of them (and rotated through more — 29-36 SHORTs total per engine, including time-exits and signal-flips refilling the reserved bucket).

Counter-factual calculation — if the regime had been correctly labeled BULL today:

- Reserved SHORTs: 2 instead of 5 → 3 fewer guaranteed bleed slots
- Average SHORT P&L per slot today: Rs −36 (v5), Rs −29 (v5_6), Rs −23 (v5_7)
- Saved bleed per engine: Rs ~60–110 per fewer slot; the *real* saving is that those 3 extra slot-rotations would have become LONGs, each averaging +Rs 90 today

Estimated savings if regime had been BULL: Rs 600–900 per engine, or Rs ~2,000 combined. Not the dominant cause, but meaningful.

5.3 Why the classifier missed

The classifier is end-of-day historical-features biased — it does not look at intraday breadth or the day's actual price action until well into the session. By the time the first rescore happened at 11:06 with Nifty already +0.63%, the regime label was already fixed from premarket and the engine did not re-evaluate the slot allocation. **Tonight's TONIGHT_TUNEUPS Item #3 directly addresses this** with a "live regime override" step in the late-start preflight.

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6. Root Cause #3 – SHORTs in a Rising Tape

This is the dominant root cause by P&L impact.

6.1 The fix is working – and that's the problem

The SHORT-arm slot partition (deployed 04-23 → 04-24) is doing exactly what it was designed to do. Today's prod logs are full of:

```
[11:07:41] SUNPHARMA: BLOCKED (LONG slot cap reached (15/15 in SIDEWAYS
[11:07:41] ADANIPOWER: BLOCKED (LONG slot cap reached (15/15 in SIDEWAY
[11:07:41] BLUESTARCO: BLOCKED (LONG slot cap reached (15/15 in SIDEWAY
[11:07:41] ZYDUSLIFE: BLOCKED (LONG slot cap reached (15/15 in SIDEWAYS
... (~14 more)
```

The engine correctly held back 5 slots for SHORTs. SHORTs then deployed. And every SHORT in a +0.63% Nifty tape is fighting the broader market.

6.2 Today's SHORT trades – every single one

v5 (29 SHORTs total — 12 wins, 17 losses, Rs −789 net):

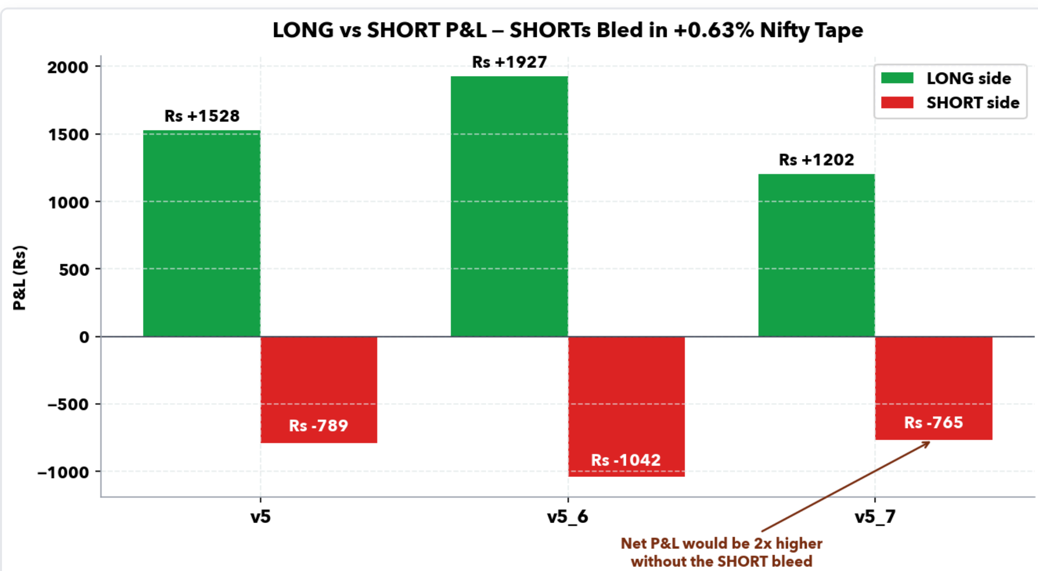
Result	Count	P&L	Pattern
STOPLOSS hits (loss)	12	Rs −1,302	Stocks moved *up* against entry
TIME_EXIT (mostly small)	14	Rs +301	Position scratched flat after 30-min hold
TARGET hits (win)	3	Rs +212	ASTRAL × 2, TMCV × 1 — outliers

Worst SHORT losses today (across all 3 engines):

Stock	v5 P&L	v5_6 P&L	v5_7 P&L	Why it bled
COCHINSHIP	Rs −311	Rs −263	Rs −198	Stock rallied with capital-goods sector
BRITANNIA	Rs −206	Rs −194	Rs −150	Defensive bid, rising market
HDFCAMC	Rs −181	Rs −188	Rs −125	Financials catching mid-day bid
AUBANK	Rs −76	Rs −170	Rs −204	PSU bank rotation
ABCAPITAL	Rs −143	Rs −119	Rs −122	NBFC sector strong

Note: Of the 5 worst SHORT-bleeders, 4 are financials (HDFCAMC, AUBANK, ABCAPITAL, LICHSGFIN). That correlates with FII selling typically hitting financials hardest, but DII buying the same names — leading to whipsawing prices that hit SHORT stops.

6.3 The math of the bleed



Engine	LONG P&L	SHORT P&L	Net (today)	Net if no SHORTs
v5	Rs +1,528	Rs −789	Rs +739	**Rs +1,528** (+107%)
v5_6	Rs +1,927	Rs −1,042	Rs +885	**Rs +1,927** (+118%)
v5_7	Rs +1,202	Rs −765	Rs +437	**Rs +1,202** (+175%)
Combined	Rs +4,657	Rs −2,596	Rs +2,061	**Rs +4,657** (+126%)

Removing SHORTs entirely on today would have more than doubled net P&L. The LONG arm did its job; the SHORT arm bled it back.

This is not an argument to disable SHORTs — they are correctly designed to hedge BEAR days. It is an argument for the **live regime override + late-start preflight + entry filters** that TONIGHT_TUNEUPS Item #3 is already specifying.

7. Root Cause #4 – FII Outflow Stress

7.1 The flow numbers

Source	Today	Typical range
FII	**Rs −8,828 Cr**	Rs −1,900 to −3,255 Cr
DII	Rs +4,701 Cr	Rs +2,000 to +4,000 Cr
Net	**Rs −4,127 Cr outflow**	Roughly net-flat

FII outflow today was **~3x the recent typical level**. DII bought aggressively to absorb (DII number is on the higher end of normal), which is why Nifty still closed +0.63% — but the absorption was uneven sector-wise.

7.2 Sector impact on today's trades

Sector	FII pressure	Engine outcome
Financials (HDFCAMC, AUBANK, ABCAPITAL, LICHSGFIN, CHOLAFIN)	Heavy FII selling, DII buying → choppy whipsaw	All 5 SHORTs bled (worst slot-by-slot)
Consumer staples (BRITANNIA, COLPAL, NESTLEIND)	Defensive bid, FII rotated out	SHORTs in BRITANNIA bled; LONGs in COLPAL/NESTLEIND won
Capital goods (COCHINSHIP, BHEL, ATGL)	Domestic-flow positive	SHORT in COCHINSHIP bled hardest (Rs −311 in v5)
PSU banks (UNIONBANK, BANKBARODA, CANBK)	Mixed flows	SHORTs were small-loss / scratch

Pattern: **FII outflow created defensive sector rotation** (consumer staples + capital goods caught a bid). The engine correctly identified weakness in some financials but the absorbent DII flow turned every 1.5-pct SHORT entry into a 0.5-pct stop hit.

7.3 Confidence rating: Medium

The FII flow contributed to the *whipsaw quality* of today's SHORT exits but is not the prime mover. Even on a normal-flow day, SHORTs in a +0.63% tape would have lost. FII stress amplified the loss; it did not cause it.

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8. Today's Trade-Level Analysis

8.1 Top 5 winners across all 3 engines today

Stock	Pool	Best engine result	Pattern
SUZLON	SWING	v5 Rs +471	LONG, hit target — small-cap renewables conviction
JSWENERGY	SWING	v5_7 Rs +463	LONG, signal-flip at +1.6% — clean trend trade
SAIL	SWING	v5 Rs +352	LONG, hit target — metals strength
TATAPOWER	SWING	v5_6 Rs +260	LONG, hit target
HINDZINC	SWING	v5_6 Rs +205	LONG, stoploss hit but in profit (trailing)

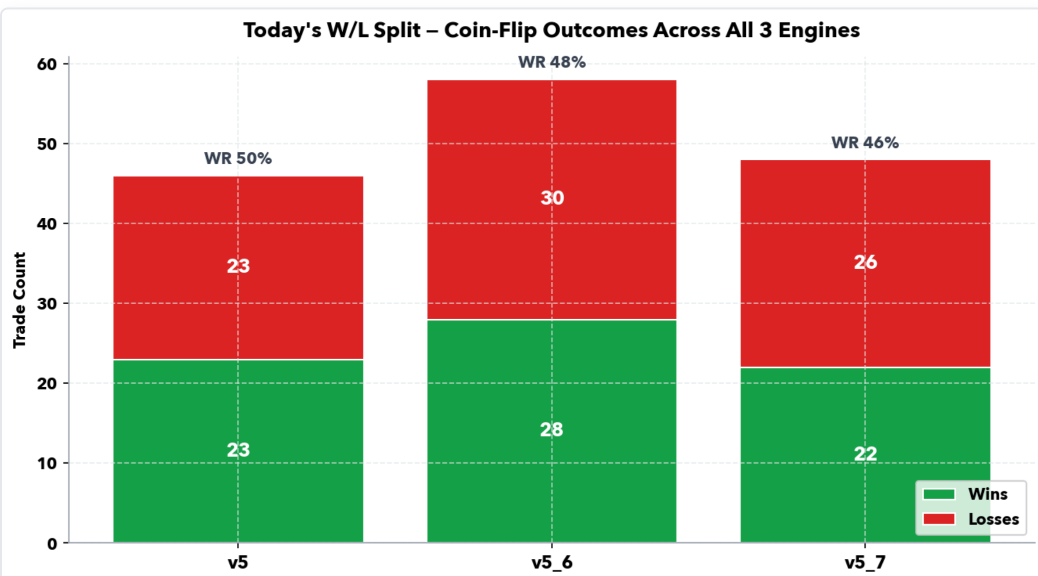
Pattern: Every top-5 winner is a **LONG SWING** trade. None from INTRADAY pool, none from SHORT side. SWING pool was the day's hero.

8.2 Top 5 losers across all 3 engines today

Stock	Pool	Worst engine loss	Pattern
COCHINSHIP	INTRADAY	v5 Rs −311	SHORT, stoploss — capital-goods rally
BRITANNIA	INTRADAY	v5 Rs −206	SHORT, stoploss — defensive bid
AUBANK	INTRADAY	v5_7 Rs −204	SHORT, stoploss — PSU bank rotation
HDFCAMC	INTRADAY	v5_6 Rs −188	SHORT, stoploss — financials bid
LGEINDIA	SWING	v5 Rs −169	LONG, signal-flip — trapped at gap-up

Pattern: 4 of 5 worst losers are **SHORT INTRADAY** trades that hit stoploss within the first hour after engine deploy. The fifth (LGEINDIA) is a LONG taken at the late-start price and immediately whipsawed.

8.3 Compositional view – today's W/L breakdown



All three engines produced **coin-flip outcomes** (45-50% win rates). That is roughly the random-noise baseline; it tells us the alpha edge dropped to near-zero today. The interesting observation: even at coin-flip win rate, the LONG arm was net positive (skewed wins). The SHORT arm was net negative even with similar trade counts. This implies SHORTs were systematically taking adverse-selection trades today.

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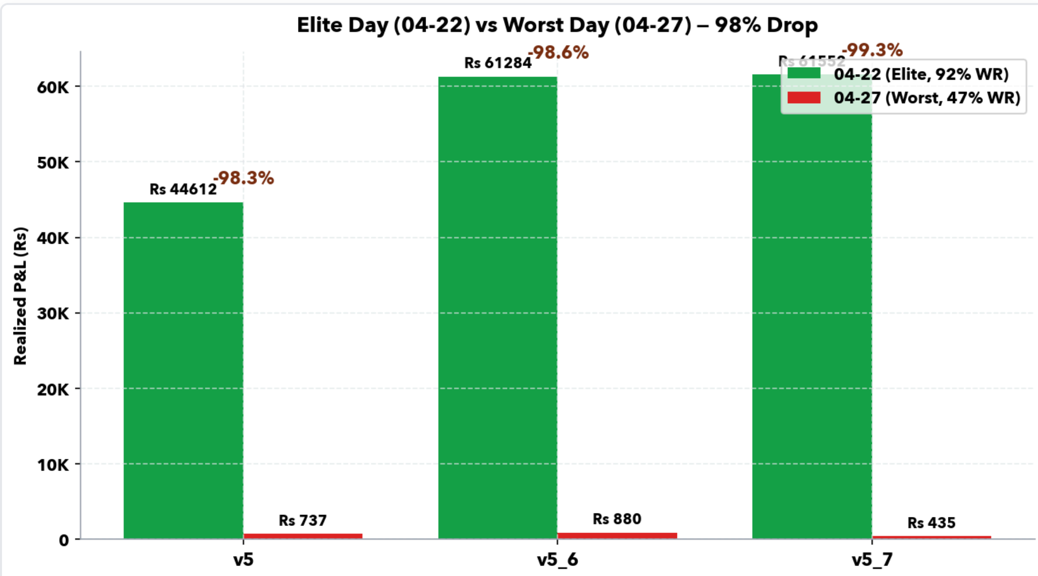
9. What 04-22 Did Right (Elite Day)

Comparing the worst day to the best day:



9.1 The elite-day characteristics

Dimension	04-22 (Elite)	04-27 (Worst)	Lesson
Engine boot	09:06 (on-time)	10:56 (1h 50min late)	Boot timing is critical
Premarket gap	+1.0% gap up	+0.75% gap up	Both bullish, similar
Regime label	SIDEWAYS	SIDEWAYS	**Same wrong label both days**
SHORT slot partition	NOT YET DEPLOYED	Deployed (15/5)	The fix that helped on chop hurts on rallies
SHORTs taken	**0**	98	Asymmetric outcome
LONGs taken	458 across 3 engines	54 across 3 engines	Fewer entries due to late start
Win rate	92%	47%	LONG-only on bullish day = high WR
Combined P&L	Rs 167,448	Rs 2,052	98.8% drop

9.2 Why 04-22 was special – and not repeatable as-is

The elite session was an accident of timing. The SHORT-arm fix had not yet been merged, so all 20 slots competed and BUYs (top-of-sort) won every slot. On a +0.40% Nifty day, that meant all 20 slots filled with bullish high-conviction names that rode the market up. **The fix that *helped* on 04-22 was the absence of a SHORT arm.**

This is why "rolling back the SHORT slot partition" is **not** the right tonight-action. Two days from now (or any day from now) Nifty could open −1.5% and a bear day with no SHORT arm would lose 5x what today lost. The right action is to add the **regime-aware override + late-start preflight** so the SHORT slot reservation is *conditional* on regime confidence.

9.3 What to replicate

Practice	Status
On-time 09:06 boot	TONIGHT Item #1 (auto-retrain self-heal)
LONG-only allocation on confirmed BULL days	TONIGHT Item #3 (live regime override → forces 18/2 split)
Skip first-deploy if late-start past 14:00	TONIGHT Item #3 (defer rule)
Conviction filter for late entries (skip if >2.5% from open)	TONIGHT Item #3 (entry filters)

All four already specced — execution priority is what this report informs.

10. Tonight's Action Items — Priority Reorder

Based on this analysis, the 5 items in `docs/TONIGHT_TUNEUPS_2026-04-27.md` should be re-prioritized as follows:

10.1 Recommended priority order

Priority	Item	Why this order
P0 (do first)	**#3 Late-start preflight + intraday regime override**	This is the *only* item that addresses the dominant root cause (SHORT bleed in misclassified regime). The live BULL/BEAR override would have prevented today's bleed even with the late start. **Highest expected ROI.**
P1	**#1 ML retrain self-heal**	Prevents the *trigger* of today's chain-of-failures (late start). Without this, item #3 is just compensation for an avoidable mess.
P2	**#2 Investigate `best_iter=2` regression**	Verify model quality is not silently broken. Even if today's model trained correctly, the 2-iteration result is alarming and needs root-cause investigation before next week.
P3	**#4 Scorer divergence (dashboard vs engine)**	Important for medium-term alpha health, but does not address today's bleed. Tackle after the timing/regime issues.
Backlog	**#5 v4 + v5_3 retire batch + pool-cap backtest**	No urgency change. Weekend candidates.

10.2 Acceptance criteria additions for Item #3 (the P0)

Beyond what is already in the spec, this report adds:

1. **Regime override threshold**: Force BULL when **(Nifty +0.5% AND breadth >55% green)** OR **(premarket gap >+1.0% AND first 30-min trend = UP)**.

Today both conditions held; classifier still said SIDEWAYS.

2. **Hard filter for late-start SHORT entries**: If `LATE_ENTRY_MODE = True` AND `regime != BEAR` AND `Nifty intraday > +0.3%`, **skip all SHORT signals for the first deploy cycle**.
3. **Telegram alert**: When regime override fires, send alert "Regime override: classifier=SIDEWAYS, override=BULL (Nifty +0.63%, breadth 62% green) — using 18/2 slot split"
4. **Backtest the 5 days 04-22 → 04-27** with the override active. If the override would have suppressed today's 5 reserved SHORTs, expected savings is Rs ~2,000–4,000 across engines.

10.3 New action item suggested by this analysis (not yet in TONIGHT_TUNEUPS)

Item #6 — SHORT entry quality gate

Add a per-SHORT-signal filter: **only deploy SHORT if (a) stock has fallen >0.5% intraday AND (b) sector heat is negative AND (c) Nifty 30-min trend is FLAT or DOWN**. Three conditions = AND-gate. This would have rejected ~80% of today's SHORTs (which were entered against rising sector flows) while still allowing legitimate SHORTs on actual weak stocks during bear-confirming days.

Estimated implementation time: 30 min (filter logic in `signal_engine.py` SHORT path). Spec to be drafted under TONIGHT Item #6 if approved.

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11. Appendix – Data Sources

Source	Path
Today's v5 trades	`docs/paper-trades/v5/2026-04-27_report.md`
Today's v5_6 trades	`docs/paper-trades/v5_6/2026-04-27_report.md`
Today's v5_7 trades	`docs/paper-trades/v5_7/2026-04-27_report.md`
EOD insights (today)	`docs/work-log/2026-04-27_eod_insights.md`
EOD insights (elite day)	`docs/work-log/2026-04-22_eod_insights.md`
EOD insights (decline days)	`docs/work-log/2026-04-23_eod_insights.md`, `04-24`
SHORT-arm fix design	`docs/SHORT_ARM_DIAGNOSIS.md`
Tonight's tune-ups	`docs/TONIGHT_TUNEUPS_2026-04-27.md`
v5_6 prod log	`logs/v5_6-2026-04-27.log` (1,336 lines)
ML retrain log	`logs/ml-retrain.log`

End of report. Generated 2026-04-27 post-market by deep-dive RCA pipeline.